Customer Churn Analysis Report Visualization

Stage 2: Data Preparation, ANN Modelling, and Customer Segmentation

Abstract

This report presents the Stage 2 deliverables for the Customer Churn Analysis project. The work focuses on data preparation, supervised modelling support, and unsupervised customer segmentation. The dataset was preprocessed through numerical scaling and categorical transformation to support downstream modelling. An Artificial Neural Network framework was prepared for churn prediction. In addition, K-means clustering was applied to identify hidden customer segments. The clustering results suggest that customer segmentation is mainly influenced by internet service type and contract structure, while tenure and senior citizen status contribute less to cluster separation.

1 Introduction

Customer churn analysis is an important task in business analytics because retaining existing customers is often more cost-effective than acquiring new ones. This project aims to support churn prediction and customer segmentation through structured data preparation and machine learning analysis. The main objectives of this stage are to prepare a clean modelling dataset, support ANN-based classification, and identify meaningful customer groups using clustering methods.

2 Dataset Overview

The raw dataset contains customer-related attributes and a binary churn outcome. The selected original features are shown in Table 1. The target variable is `Churn`, where `Yes` indicates customer churn and `No` indicates customer retention.

Table 1: Original features used in the project

Feature	Type
gender	Categorical
SeniorCitizen	Numerical / Binary indicator
Dependents	Categorical
tenure	Numerical
PhoneService	Categorical
MultipleLines	Categorical
InternetService	Categorical
Contract	Categorical
MonthlyCharges	Numerical
Churn	Target label

3 Data Preparation

The dataset was preprocessed to ensure compatibility with ANN modelling and clustering analysis. The target variable `Churn` was mapped into binary form. Specifically, `Yes` was encoded as 1 and `No` was encoded as 0.

3.1 Numerical Features

The numerical features included:

- `SeniorCitizen`
- `tenure`
- `MonthlyCharges`

These features were transformed using `MinMaxScaler`. This process normalised the values into a comparable range and improved suitability for ANN training and distance-based clustering.

3.2 Categorical Features

The categorical features included:

- `gender`
- `Dependents`
- `PhoneService`
- `MultipleLines`
- `InternetService`
- `Contract`

Among these variables, `Contract` was transformed through one-hot encoding. The remaining categorical variables were converted into numerical codes for modelling.

Table 2: Feature transformation summary

Feature	Transformation	Interpretation
Churn	Binary mapping	0 = No churn, 1 = Churn
SeniorCitizen	MinMax scaling	Binary numerical indicator where 0 = Not senior citizen, 1 = Senior citizen
tenure	MinMax scaling	Continuous variable representing customer tenure
MonthlyCharges	MinMax scaling	Continuous variable representing monthly billing amount
gender	Category encoding	Encoded from original categories Female and Male
Dependents	Category encoding	Encoded from original categories No and Yes
PhoneService	Category encoding	Encoded from original categories No and Yes
MultipleLines	Category encoding	Encoded from original categories No and Yes
InternetService	Category encoding	Encoded from original categories such as DSL and Fiber optic
Contract	One-hot encoding	Expanded into `Contract_Month-to-month` and `Contract_One year` according to the observed dataset categories

3.3 Train and Test Preparation

For supervised modelling, the processed dataset was split into training and testing subsets using an 80/20 ratio. This split was applied with a fixed random state to maintain reproducibility. The processed data supports both ANN-based churn prediction and clustering-based customer segmentation.

4 Artificial Neural Network Support

After preprocessing, the cleaned dataset was used to support ANN model development. The ANN structure included an input layer, two hidden dense layers, and a sigmoid output layer for binary classification. The network was trained using the Adam optimiser and binary cross-entropy loss. This setup provides a suitable baseline model for churn prediction.

The ANN workflow included:

- feature-label separation
- train-test split
- model training
- prediction on the test set
- confusion matrix and classification report generation

This modelling stage demonstrates that the prepared dataset is ready for supervised learning tasks.

5 Customer Segmentation with K-means

In addition to churn prediction, clustering was applied to explore whether the customer base could be segmented into distinct groups. K-means clustering was selected because it is interpretable and suitable for structured tabular data. The elbow method was used to determine a suitable number of clusters, and PCA was applied for two-dimensional visualisation.

5.1 Elbow Method

Figure 1 presents the elbow method results. The inertia decreases sharply when moving from one cluster to four clusters. After that point, the rate of improvement becomes more gradual. This pattern suggests that $k = 4$ is a reasonable choice because it balances segmentation detail and model simplicity.

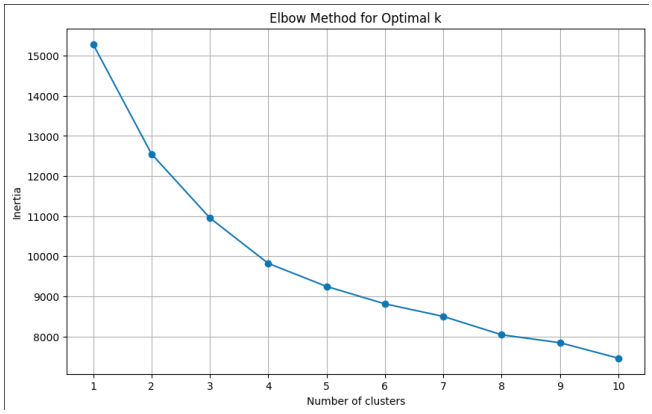

Figure 1: Elbow method for selecting the optimal number of clusters

5.2 PCA Cluster Visualisation

Figure 2 shows the clustering output projected onto two principal components. The four clusters appear visually separated with limited overlap. This indicates that the K-means model successfully identified distinct customer segments in the transformed feature space.

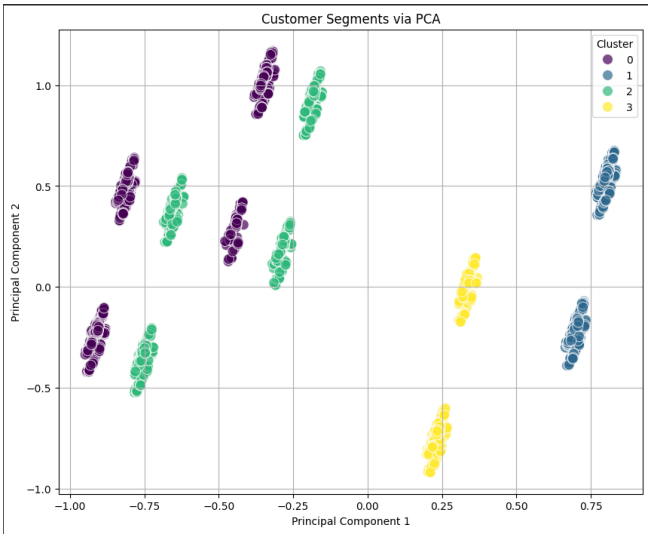

Figure 2: Customer segments visualised through PCA

5.3 Key Features Driving Segmentation

The most important variables contributing to segmentation are shown in Table 3. Internet service type was the strongest driver of cluster separation. Contract-related features also contributed substantially. In contrast, `tenure` and `SeniorCitizen` had no visible contribution in this output. This indicates that the clustering pattern is driven more by service choice and contractual structure than by duration of relationship or age-based demographic status.

Table 3: Top features driving segmentation

Feature	**Importance**
InternetService	0.365431
Contract_Month-to-month	0.334377
Contract_One year	0.300192
tenure	0.000000
SeniorCitizen	0.000000

5.4 Cluster Distribution Analysis

Figure 3 summarises the cluster-level distributions of tenure and monthly charges. The tenure boxplots are broadly similar across the four clusters. This supports the feature importance output because tenure does not appear to be a major source of separation.

Monthly charges show moderate variation across clusters. However, the differences are less pronounced than those associated with internet service and contract structure. Therefore, monthly billing level may play a secondary role, while service type and contract pattern remain the dominant segmentation drivers.

6 Discussion

The clustering analysis provides several useful business insights. First, internet service type is the strongest factor influencing customer segmentation. This suggests that customers using different service offerings may have distinct behaviours or service needs. Second, contract structure also plays a major role. Month-to-month and one-year customers appear to form different groups, which may reflect differences in commitment level and churn risk.

By contrast, tenure does not separate the clusters in a meaningful way. The boxplots and feature importance values both support this conclusion. Senior citizen status also contributed little in this clustering output. Therefore, demographic indicators alone may be insufficient for segmenting these customers.

These findings imply that targeted retention strategies should focus more on service configuration and contract terms. Segment-specific interventions may be more effective than a single general retention policy.

7 Conclusion

This report summarised the Stage 2 work for customer churn analysis. The dataset was prepared through structured preprocessing, including binary mapping, numerical scaling, categorical encoding, and one-hot transformation. The processed data supported ANN model development for churn prediction. In parallel, K-means clustering identified four meaningful customer segments.

The results show that customer segmentation is primarily influenced by internet service type and contract structure. Tenure and senior citizen status were less useful for distinguishing groups in this analysis. Overall, the project demonstrates that a clean and well-structured data preparation pipeline can effectively support both predictive modelling and exploratory segmentation.

Appendix

All figures used in this report should be placed in the same folder as the `main.tex` file:

- `Elbow.png`
- `PCA.png`
- `Cluster.png`

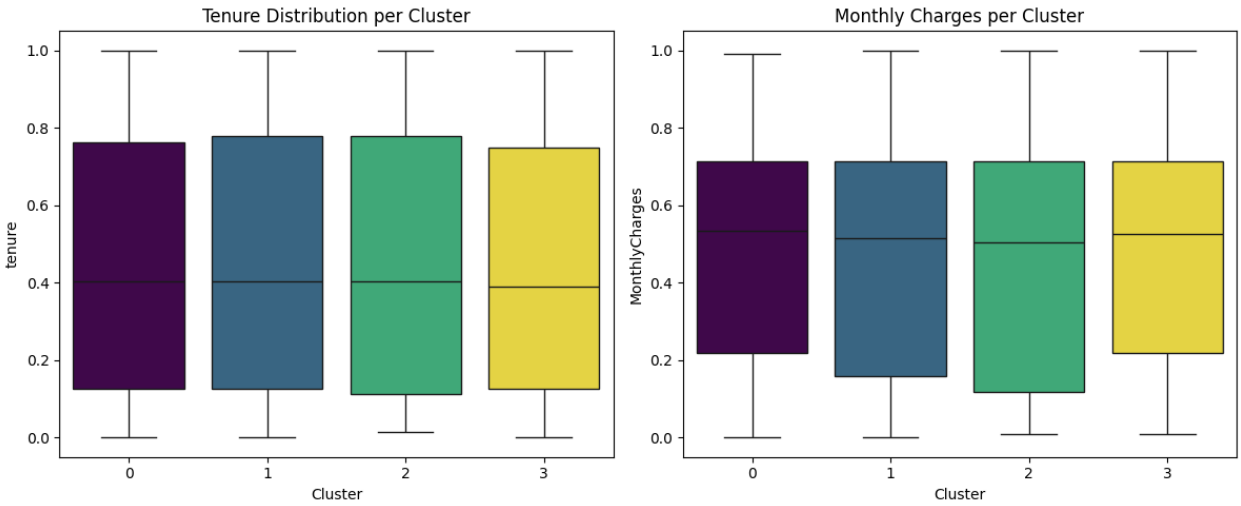

Figure 3: Cluster interpretation through feature importance and distribution analysis

Table 4: Interpretation of clustering results

Cluster	**Observed pattern**	**Main drivers**	**Interpretation**
0	Distinct position in PCA space with similar tenure spread to other clusters	InternetService and Contract	Represents a service and contract combination that differs clearly from the other groups
1	Clearly separated in PCA space and located on the positive side of the first principal component	InternetService and Contract	Likely corresponds to customers with a different service profile and contract structure from Cluster 0
2	Separated from Clusters 0 and 1 with comparable tenure dispersion	InternetService and Contract	Indicates another customer segment shaped mainly by service type and billing arrangement
3	Compact and distinguishable cluster with moderate monthly charge variation	InternetService and Contract	Represents a smaller but clearly defined segment within the customer base